Pranav Srinivasan

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EDUCATION

Kent State University Kent, OH
Bachelor of Science in Physics *Expected May 2027*

- **Relevant Coursework:** Thermodynamics, Quantum Mechanics, Control Systems, MATLAB Programming

RESEARCH EXPERIENCE

Undergraduate Researcher Oct 2023 – Present
Dr. Thorsten Schmidt Lab, Kent State University *Kent, OH*

- Investigated DNA origami structures, focusing on support structures to suspend DNA nanodiscs for use in membrane protein imaging and structure characterization.
- Utilized CaDNAno to design DNA origami structures. Developed skills in micropipetting, DNA origami synthesis, and analysis using gel electrophoresis and Transmission Electron Microscopy (TEM).

ENGINEERING PROJECTS

HAVOC Flight Computer Aug 2025 – Present
Embedded Engineer / Contributor

- Designed and routed custom PCB boards for various flight computer subsystems.
- Integrated environmental sensor suites (BMI088, MS5611, BME688, SHT41) for high-altitude telemetry collection.

IREC High-Power Rocketry Aug 2024 – Present
Club-Side Payload Lead / Contributor *Kent, OH*

- Lead payload development and testing of the PCBs and electronics for a 4U, weather station payload driven by a Raspberry Pi and Arduino.
- Developed the software stack to interface with the sensors and control the payload's landing legs mechanism. Analysis of environmental data on-board the Raspberry Pi to calculate atmospheric conditions including wind speed and direction.

Amateur Radio & Remote Telemetry Beacon Aug 2025 – Present
Hardware & Software Implementation *Kent, OH*

- Configured a Raspberry Pi 3A+ running a Python-based telemetry stack for AX.25 packet decoding and APRS transmission.
- Interfaced HackRF and RTL-SDR hardware with SDR++ for real-time digital communications and signal analysis.
- Developed feed-horna for a repurposed WiFi dish to downlink MetOp and NOAA GOES satellite image data.

AstroFlashes, FlashSat-1 Oct 2025 – Present
Deputy Lead, Communications *Kent, OH*

- Lead the communications team for the AstroFlashes project, managing the on-board telemetry and APRS transmission.
- Developed F Prime (F') stack for on-board telemetry and APRS transmission.
- Laid the groundwork for ground station development at Andrew Paton Airport
- Helped development of a buildroot operating system for the BeagleBone Black Industrial
- Development of Student Research and Developed (SRAD) PCB carrier boards for the BeagleBone Black Industrial and Raspberry Pi Compute Module 5

TECHNICAL SKILLS

Languages: C, C++, Python, LaTeX, Bash
Hardware & Embedded: KiCad, Arduino, Raspberry Pi, BeagleBone, SDR, Oscilloscopes
Software & Tools: Arch Linux (Hyprland), Fedora, Ubuntu, Neovim, F Prime (F'), Git